

Discussion 8 Solutions

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October 2025

Formulas:

1. **Power Rule:** $\frac{d}{dx}(x^n) = nx^{n-1}$
2. **Derivative of Sine:** $\frac{d}{dx}(\sin x) = \cos x$
3. **Derivative of Cosine:** $\frac{d}{dx}(\cos x) = -\sin x$
4. **Product Rule:** $\frac{d}{dx}(u \cdot v) = u'v + uv'$
5. **Quotient Rule:** $\frac{d}{dx}\left(\frac{u}{v}\right) = \frac{u'v - uv'}{v^2}$
6. **Chain Rule:** $\frac{d}{dx}(f(g(h(x)))) = f'(g(h(x))) \cdot g'(h(x)) \cdot h'(x)$
7. **Gradient of a Function:** $\nabla f(x, y, z) = \left(\frac{\partial f}{\partial x}, \frac{\partial f}{\partial y}, \frac{\partial f}{\partial z}\right)$
8. **Directional Derivative:** $D_{\mathbf{u}}f(x, y, z) = \nabla f(x, y, z) \cdot \mathbf{u} = |\nabla f(x, y, z)| |\mathbf{u}| \cos \theta$
9. **Second Order Partial Derivative Test:** Let $D = f_{xx}(x_0, y_0)f_{yy}(x_0, y_0) - (f_{xy}(x_0, y_0))^2$.
 - If $D > 0$ and $f_{xx} > 0$, then (x_0, y_0) is a local minimum.
 - If $D > 0$ and $f_{xx} < 0$, then (x_0, y_0) is a local maximum.
 - If $D < 0$, then (x_0, y_0) is a saddle point.
 - If $D = 0$, the test is inconclusive.

Problem 1: Find the first and second order derivatives of the given function

$$f(x) = x^2 \cos(x) + \frac{2}{x}, \quad x \neq 0.$$

Solution

First derivative. Using the product rule on $x^2 \cos x$ and the power rule on $2/x = 2x^{-1}$,

$$\begin{aligned} f'(x) &= \frac{d}{dx}(x^2 \cos x) + \frac{d}{dx}(2x^{-1}) \\ &= (2x \cos x - x^2 \sin x) - 2x^{-2} \\ &= 2x \cos x - x^2 \sin x - \frac{2}{x^2}. \end{aligned}$$

Second derivative. Differentiate $f'(x)$ term-by-term:

$$\begin{aligned} f''(x) &= \frac{d}{dx}(2x \cos x) - \frac{d}{dx}(x^2 \sin x) + \frac{d}{dx}(-2x^{-2}) \\ &= (2 \cos x - 2x \sin x) - (2x \sin x + x^2 \cos x) + 4x^{-3} \\ &= (2 - x^2) \cos x - 4x \sin x + \frac{4}{x^3}. \end{aligned}$$

Problem 2: Given $f(x) = ((x^2)^2)^2$. Find $f'(x)$ using chain rule.

Solution

$$\text{Let } h(x) = x^2, \quad g(u) = u^2, \quad f(v) = v^2.$$

Taking derivatives,

$$f'(v) = 2v, \quad g'(u) = 2u, \quad h'(x) = 2x.$$

By the Chain Rule,

$$\frac{d}{dx}(f(g(h(x)))) = f'(g(h(x))) \cdot g'(h(x)) \cdot h'(x).$$

Substituting,

$$f'(x) = 2(g(h(x))) \cdot 2(h(x)) \cdot 2x.$$

Simplifying,

$$f'(x) = 8x h(x) g(h(x)).$$

Since $h(x) = x^2$ and $g(h(x)) = (x^2)^2 = x^4$,

$$f'(x) = 8x(x^2)(x^4) = 8x^7.$$

Problem 3:

Given

$$f(x, y) = x^3 + y^3 - 3x^2 + 3y^2,$$

find all the critical points and determine whether they are saddle points, local maxima, or local minima using the second order partial derivative test.

Solution**Step 1:** Find the first-order partial derivatives.

$$f_x = \frac{\partial f}{\partial x} = 3x^2 - 6x, \quad f_y = \frac{\partial f}{\partial y} = 3y^2 + 6y.$$

Step 2: Set $f_x = 0$ and $f_y = 0$ to find critical points.

$$3x^2 - 6x = 0 \quad \Rightarrow \quad x(x - 2) = 0 \quad \Rightarrow \quad x = 0 \text{ or } 2,$$

$$3y^2 + 6y = 0 \quad \Rightarrow \quad y(y + 2) = 0 \quad \Rightarrow \quad y = 0 \text{ or } -2.$$

Thus, the critical points are:

$$(0, 0), \quad (0, -2), \quad (2, 0), \quad (2, -2).$$

Step 3: Compute the second-order partial derivatives.

$$f_{xx} = 6x - 6, \quad f_{yy} = 6y + 6, \quad f_{xy} = 0.$$

Step 4: Use the second-order partial derivative test.

$$D = f_{xx}f_{yy} - (f_{xy})^2 = (6x - 6)(6y + 6).$$

Evaluate at each critical point:

(x, y)	f_{xx}	f_{yy}	D	Classification
$(0, 0)$	-6	6	$(-6)(6) = -36$	Saddle point
$(0, -2)$	-6	-6	$(-6)(-6) = 36, f_{xx} < 0$	Local maximum
$(2, 0)$	6	6	$(6)(6) = 36, f_{xx} > 0$	Local minimum
$(2, -2)$	6	-6	$(6)(-6) = -36$	Saddle point

Problem 4: The temperature T (in $^{\circ}\text{C}$) at a point (x, y, z) in space is given by the function:

$$T(x, y, z) = 4x^2y - 2y^2z + xyz$$

- Find the rate of change of temperature at the point $(1, -2, 3)$ in the direction of the vector $(2, 1, -1)$.
- In which direction does the temperature increase most rapidly at the point $(1, -2, 3)$?

Solution

Step 1: Compute the gradient of $T(x, y, z)$.

$$\nabla T = \left(\frac{\partial T}{\partial x}, \frac{\partial T}{\partial y}, \frac{\partial T}{\partial z} \right)$$

$$\frac{\partial T}{\partial x} = 8xy + yz, \quad \frac{\partial T}{\partial y} = 4x^2 - 4yz + xz, \quad \frac{\partial T}{\partial z} = -2y^2 + xy.$$

Step 2: Evaluate the gradient at $(1, -2, 3)$.

$$\frac{\partial T}{\partial x}(1, -2, 3) = 8(1)(-2) + (-2)(3) = -16 - 6 = -22,$$

$$\frac{\partial T}{\partial y}(1, -2, 3) = 4(1)^2 - 4(-2)(3) + (1)(3) = 4 + 24 + 3 = 31,$$

$$\frac{\partial T}{\partial z}(1, -2, 3) = -2(-2)^2 + (1)(-2) = -8 - 2 = -10.$$

Thus,

$$\nabla T(1, -2, 3) = \langle -22, 31, -10 \rangle.$$

Step 3: Find the unit vector in the direction of $(2, 1, -1)$.

$$\mathbf{u} = \frac{1}{\sqrt{2^2 + 1^2 + (-1)^2}} \langle 2, 1, -1 \rangle = \frac{1}{\sqrt{6}} \langle 2, 1, -1 \rangle.$$

Step 4: Rate of change of temperature in the given direction.

$$D_{\mathbf{u}}T = \nabla T \cdot \mathbf{u} = \frac{1}{\sqrt{6}} [(-22)(2) + (31)(1) + (-10)(-1)].$$

$$D_{\mathbf{u}}T = \frac{1}{\sqrt{6}} [-44 + 31 + 10] = \frac{-3}{\sqrt{6}}.$$

Hence, the rate of change of temperature is

$$D_{\mathbf{u}}T = \frac{-3}{\sqrt{6}} \text{ } ^{\circ}\text{C/unit distance}.$$

Step 5: Direction of maximum increase.

The temperature increases most rapidly in the direction of the gradient:

Direction of maximum increase: $\nabla T(1, -2, 3) = \langle -22, 31, -10 \rangle$.